THE CATEGORY OF DIFFERENTIAL GRADED MODULES IS ABELIAN AND COMPLETE AND COCOMPLETE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the abelian and colimit claims proceed by creation through the forgetful functor (the source states the commutation of kernels, cokernels, sums and colimits with the forgetful functor); the product computation (prod N_i)^n = prod N_i^n with total object the direct sum of degrees is the substantive point and is displayed in the span; equalizers are covered since kernels are equalizers; the limits criterion is standard external tooling (stub).

Editorial note. The label lemma-dgm-abelian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The category of graded A-modules. Standard background: the category of (graded) A-modules is abelian with arbitrary limits and colimits. The creation argument transfers the abelian structure and the (co)limits from the category of A-modules through the forgetful functor.
- Limits criterion. Categories, Lemma categories-lemma-limits-products-equalizers (standard tooling). Final step (source lines 435-437): a category with products and equalizers has all limits.

Source reference. Original lemma 002N: sources/stacks-chow/tags.txt, source lines 104–104. Prerequisite pointer invoked at proof line(s) 436; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let (A, d) be a differential graded algebra. The category $Mod_{(A,d)}$ is abelian and has arbitrary limits and colimits.*

Proof. Kernels and cokernels commute with taking underlying A-modules. Similarly for direct sums and colimits. In other words, these operations in $\mathrm{Mod}_{(A,\mathrm{d})}$ commute with the forgetful functor to the category of A-modules. This is not the case for products and limits. Namely, if N_i, $i \in I$ is a family of differential graded A-modules, then the product $\prod N_i$ in $\mathrm{Mod}_{(A,\mathrm{d})}$ is given by setting $(\prod N_i)^n = \prod N_i^n$ and $\prod N_i = \bigoplus_n (\prod N_i)^n$. Thus we see that the product does commute with the forgetful functor to the category of graded A-modules. A category with products and equalizers has limits, see Categories, Lemma 002N. □

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